

Blue highlight = must-remember · **IMPORTANT** = asked in past papers · **BOOK p.xxx ↗** = click to open the textbook at that page · Orange box = conflict/trap between sources.

PART B · Answer Outlines for Likely 8-Mark Questions

Each (a) and (b) is 4 marks: write 6–8 crisp points, one diagram or table if relevant. Order below follows unit order.

Question	Outline (what to write)
U1 Vedic corpus; Vedāṅgas as supporting sciences (MQP 1a)	Define Veda (vid), Śruti, 4 Vedas with features → 4 divisions → 6 Vedāṅgas with body analogy and one line each → conclude: Vedāṅgas help in recitation, interpretation, ritual (Śikṣā-Chandas-Vyākaraṇa-Nirukta-Kalpa-Jyotiṣa).
U1 Holistic IKS vs reductionist West (1b)	Table: holistic/experiential/community/oral/sustainable vs analytic/empirical/reductionist/documented/control of nature; examples Ayurveda vs allopathy; need for synthesis.
U1 Historicity of IKS (SEE 1b)	Vedic period → epics/classical → philosophy → maths/astronomy/medicine → Nalanda/Takshashila → manuscripts → colonial disruption → NEP 2020 revival.
U1 TKDL and biopiracy (MQP 2b)	Define biopiracy → turmeric/neem cases → TKDL (database, prior art, patent offices) → outcome.
U2 Pāṇini as computational grammar (MQP 3a / SEE 3a)	Aṣṭādhyāyī facts (3,983, 8×4) → root + suffix + rule → recursion/anuvṛtti → CFG/BNF table → example Rāmaḥ phalam khādati → generative power (new words) → NLP/AI relevance.
U2 Piṅgala & binary / de Bruijn (MQP 3b / SEE 4b)	Laghu/Guru and 4 guru types → L=1/G=0 → 8 gaṇas table → mnemonic and binary cycle → de Bruijn → prastāra, naṣṭa, uddiṣṭa, meru-prastāra, Pascal.
U2 Knowledge triangle (MQP 4a)	Pramātṛ-Prameya-Pramāṇa with figure → Vaiśeṣika (6 padārthas) and Nyāya (4 pramāṇas) → Saṃśaya starts inquiry → tarka → siddhānta → nirṇaya.
U2 Origin of zero (MQP 4b); Decimal place value (SEE 3b)	Evidence timeline: Piṅgala śūnya → Bakhshālī dot → Āryabhaṭa place value → Brahmagupta 628 CE rules → inscriptions 594/876 CE → Arabs (Al-Khwārizmī) → Europe (Fibonacci). Advantages vs Roman numerals.
U2 Anumāna with 5 steps (SEE 4a)	Define anumāna (hetu, vyāpti, pakṣa, sādhyā) → 5-step table with the ball or smoke/fire example.
U3 Śulba Sūtras: ritual → geometry (MQP 5a)	Altar construction needs exact shapes → rope constructions → Baudhāyana theorem → triples, $\sqrt{2}$, π → shape transformations → shift from ritual to reasoning.
U3 Āryabhaṭa's astronomy (MQP 5b)	Āryabhaṭīya structure → rotation of Earth (boat analogy) → eclipses by shadow, Moon's reflected light → sine, π → manda/śīghra → influence on later astronomers/Arabs.
U3 Pañchāṅga (MQP 6a / SEE 6a)	Five limbs table → basis (Sun–Moon motion, 12°, 27 nakṣatras) → adhika māsa → uses (muhūrta, festivals, farming) → science + culture.
U3 Wootz (MQP 6b); Delhi Pillar (SEE 6b)	See starred boxes: process, properties, Damascus, decline; pillar facts and passive layer.
U3 Four mathematicians (SEE 5a); perfumes (SEE 5b)	Choose Āryabhaṭa, Brahmagupta, Bhāskara II, Mādhava/Piṅgala with work and contribution; perfume: sources, deg-bhapka, fixatives, uses, trade.

U4 Tridoṣa physical + psychological (MQP 7a); Pitta (SEE 7a)	Elements and functions table → prakṛti/vikṛti → aggravation signs → link to guṇas (psychological) → balancing.
U4 64 Kalās (MQP 7b / SEE 8a)	Meaning, source (Kāma Sūtra), categories table → social/economic life (skills, crafts, trade) → modern holistic education.
U4 Irrigation as sustainable water mgmt (MQP 8a)	Harappan reservoirs → Ahar-Pyne → tanks, Kallanai → stepwells, johads → community upkeep → modern relevance.
U4 Architecture: functionality + aesthetics (MQP 8b); Nāgara vs Drāviḍa (SEE 7b)	Comparison table + examples; geometry, acoustics, astronomy, materials; symbolism. Vāstu in modern design (SEE 8b).
U5 Safeguarding TK in globalisation (MQP 9a)	Threats → methods → TKDL, GI, BDA → turmeric/neem → importance.
U5 Dharma-based governance of Rāmāyaṇa (MQP 9b); advisors (SEE 9a)	Principles (citizen-centric, councils, ethical law, crisis, sustainability) + modern table; Vasiṣṭha and ministers.
U5 Traditional governance → sustainable development (MQP 10a/b)	Saptāṅga, Kośa, lokakalyāṇa, IKS–SDG table, integration with modern frameworks (policy, education, local self-reliance).

Last checks before you sleep

- Conventions: **Laghu = 1, Guru = 0**. Digits in saṃkhyā systems are read **right to left**. Sūtras = **3,983**. Pañchāṅga ≠ Rāśi.
- Conflicts flagged above (Vāstu limbs, π value, Piṅgala's date, khādri-rāmāgnayah) are exactly where marks are lost: keep the stated answer.
- If a textbook link does not jump, open the textbook PDF yourself and go to PDF page = printed page $\div 2 + 17$ (rounded down).

All the best. Prepared from your own course material; a few textbook-sourced extras (irrigation, Kośa revenue, Prameya detail, Piṅgala algorithms) are marked so you know where they come from.